

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – Industry Problem Solving

|                         |                                                                                                          |                      |                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------|
| <b>Course Code:</b>     | <b>DSA0404</b>                                                                                           | <b>Course Title:</b> | <b>FUNDAMENTALS OF DATA SCIENCE</b>                |
| <b>CO Assessed:</b>     | CO4-AT2 Industry Problem Solving                                                                         | <b>Assessment:</b>   | <b>Assessment Tool 2- Industry Problem Solving</b> |
| <b>Weightage:</b>       | <b>40% of CIA</b>                                                                                        | <b>Total Marks:</b>  | <b>100</b>                                         |
| <b>CO4 Statement:</b>   | To apply the statistical inferences such as testing of hypothesis and point estimates for the given data |                      |                                                    |
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| <b>Section / Batch:</b> | 2023/AI&DS                                                                                               | <b>Date:</b>         | 24.08.26                                           |

### Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

| Criteria                | Weight age | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                    | Level 1 – Beginning                                  | Marks |
|-------------------------|------------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------------|-------|
| Understanding of Case   | 25         | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |       |
| Application of Concepts | 25         | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts         | Fails to apply relevant concepts                     |       |
| Analysis                | 20         | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth       | Weak or no analysis; lacks reasoning                 |       |
| Solution Design         | 20         | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification | Incorrect or no solution provided                    |       |
| Clarity                 | 10         | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure           | Poor clarity; difficult to understand                |       |

**Que:** A bank wants to estimate the average monthly transaction value of its customers. A sample of 200 customers has a mean transaction value of ₹18,500 and standard deviation of ₹4,200. Calculate the point estimates of the population mean and variance.

### **Problem Statement**

A bank wants to estimate the **average monthly transaction value** of its customers. A random sample of **200 customers** gives:

- Sample size:
- Sample mean:
- Sample standard deviation:

Calculate the **point estimates of the population mean and population variance**.

### **1. Introduction to Point Estimation**

**Point estimation** is a statistical method used to estimate an unknown population parameter using a single numerical value calculated from a sample.

In this banking problem, the bank is interested in two population parameters:

1. **Population mean** – the true average monthly transaction value of all customers.
2. **Population variance** – the true variability in monthly transaction values of all customers.

Since the bank cannot collect transaction information from every customer, it takes a random sample of 200 customers and uses the sample statistics as estimates.

### **Basic idea**

For example:  $\bar{x} \rightarrow \mu$  and  $s^2 \rightarrow \sigma^2$

### **2. Point Estimate of Population Mean**

The **sample mean** is the point estimator of the population mean.

The formula is:

$$\hat{\mu} = \bar{x}$$

Given:  $\bar{x} = ₹18,500$

Therefore,  $\mu^{\wedge} = ₹18,500$

### Interpretation

The **point estimate of the population mean monthly transaction value is ₹18,500.**

This means that, based on the sample of 200 customers, the bank estimates that the average monthly transaction value of its entire customer population is approximately **₹18,500 per customer.**

### 3. Point Estimate of Population Variance

The sample standard deviation is:  $s = ₹4,200$

The population variance is estimated using the **sample variance:**

$$\hat{\sigma}^2 = s^2$$

Therefore,

$$s^2 = (4,200)^2$$

$$s^2 = 17,640,000$$

Thus,

$$\hat{\sigma}^2 = ₹17,640,000^2$$

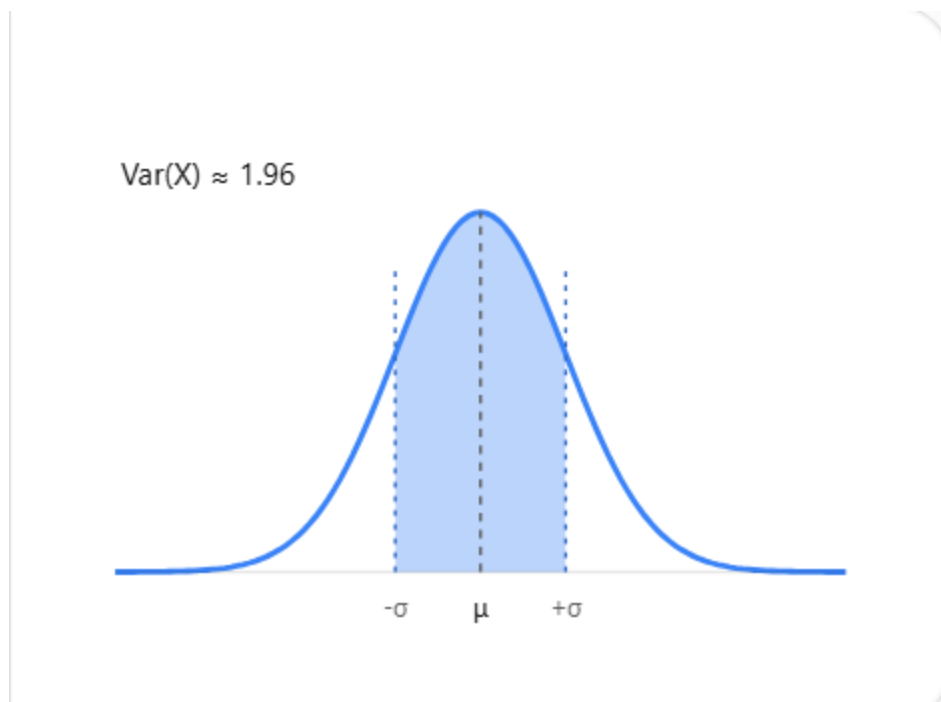
More precisely, since variance is measured in **squared rupees:**

$$\hat{\sigma}^2 = 17,640,000 \text{ rupees}^2$$

A larger variance indicates greater variation in the transaction values among customers.

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Var}(X) = (1.4)^2 = 1.96$$



#### 4. Why Are Sample Statistics Used?

The bank wants information about **all customers**, but collecting the monthly transaction value of every customer may be expensive and time-consuming. Therefore, a random sample of 200 customers is selected.

The sample provides:

$\bar{x} = ₹18,500$  and  $s = ₹4,200$

#### 5. Important Formulae

##### Population Mean

$$\hat{\mu} = \bar{x}$$

##### Population Variance

$$\hat{\sigma}^2 = s^2$$

If the sample standard deviation is given:

$$s^2 = (s)^2$$

For this problem:

$$s^2 = (₹4,200)^2 = 17,640,000 \text{ rupees}^2$$

## 6. Step-by-Step Numerical Calculation

### Step 1: Identify the sample information

$$n=200$$

$$\bar{x}=\text{₹}18,500$$

$$s=\text{₹}4,200$$

### Step 2: Estimate population mean

$$\mu^{\wedge}=\bar{x}$$

$$\mu^{\wedge}=\text{₹}18,500$$

### Step 3: Estimate population variance

$$\sigma^2=s^2$$

$$\sigma^2=(\text{₹}4,200)^2$$

$$\sigma^2=\text{₹}17,640,000 \text{ rupees}^2$$

## 7. Banking Industry Interpretation

The result can be interpreted from a banking perspective.

The bank estimates that its customers have an **average monthly transaction value of ₹18,500**.

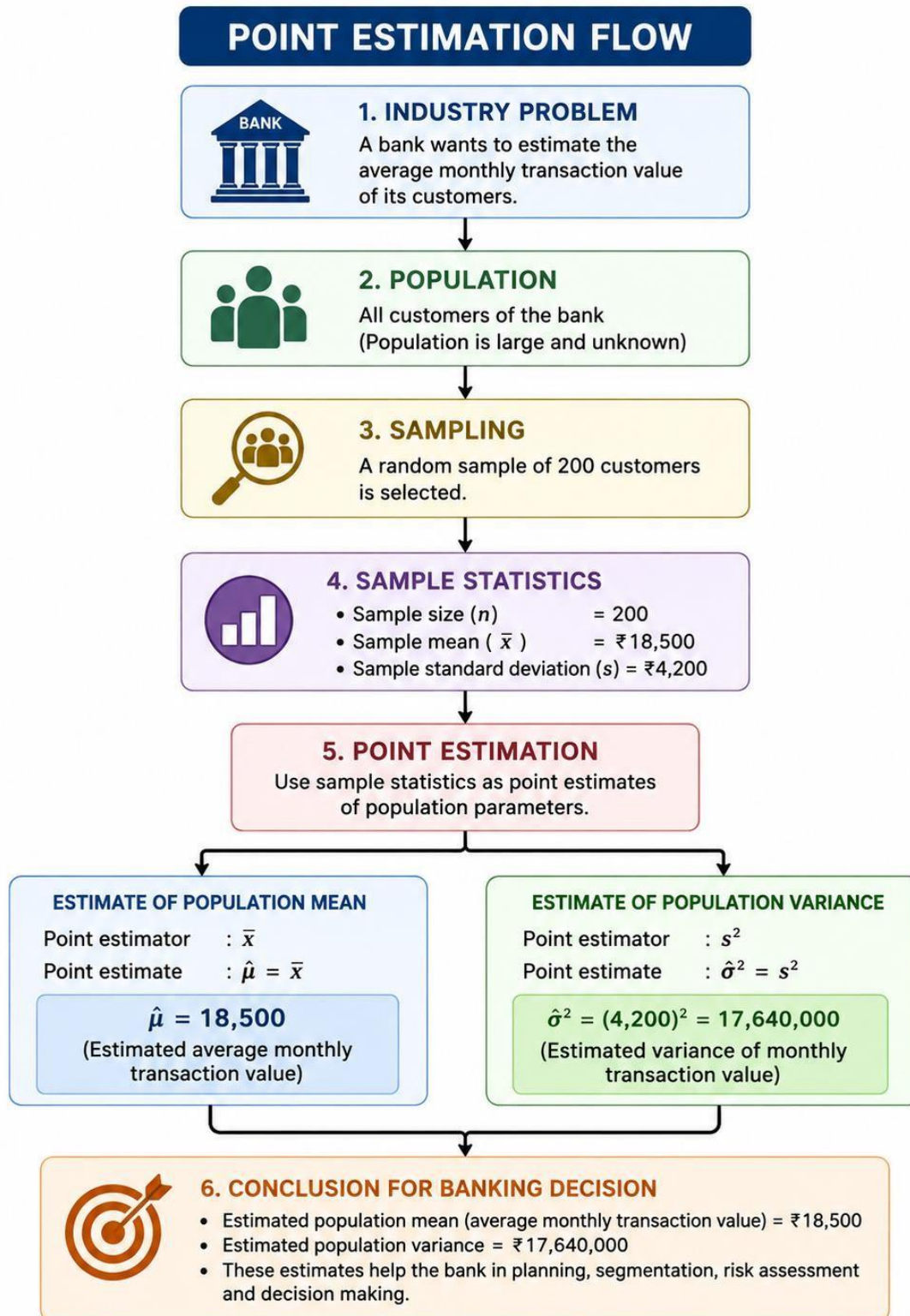
The estimated variance is: 17,640,000 rupees<sup>2</sup>

The corresponding sample standard deviation of **₹4,200** indicates that individual customers' monthly transaction values can vary considerably around the average.

This information can help the bank in:

- customer segmentation,
- transaction-volume analysis,
- financial planning,
- credit and risk assessment,
- designing banking products,
- identifying high-value customers, and
- estimating expected transaction volumes.

## 8. Point Estimation Flow



### Final Answer

| Parameter             | Formula                  | Calculation            | Point Estimate            |
|-----------------------|--------------------------|------------------------|---------------------------|
| Population Mean $\mu$ | $\mu^{\wedge} = \bar{x}$ | ₹18,500                | ₹18,500                   |
| Population Variance   | $\hat{\sigma}^2 = s^2$   | (4,200 <sup>^2</sup> ) | 17,640,000 ₹ <sup>2</sup> |

### Conclusion

Therefore, using the sample of **200 customers**, the bank's point estimates are:  
Hence, the bank can use **₹18,500 as the estimated average monthly transaction value** for its entire customer population, while **17,640,000 ₹<sup>2</sup>** represents the estimated variability of transaction values.